

CSE 3200: Software Development Project II

PIXAI: Political Image Cross Attention Inference

A Multimodal Framework for Code-Mixed Bangla Political Meme Classification

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The problem: Challenges in Meme Classification

Semantic Gap in Multimodality

Image and text analyzed in isolation miss implicit reasoning.

Linguistic Complexity

Bengali-English code-mixing ("Banglish") breaks traditional OCR systems

Visual Noise

Compression artifacts, watermarks, non-standard fonts degrade feature extraction.

Motivation: Why It Matters

Technical Gap:

- Existing models fail on **low-resource**, **code-mixed** languages
- Simple concatenation-based fusion insufficient for **nuanced reasoning**
- Traditional OCR **inadequate for noisy** meme text

Societal Impact:

- **Making computational models understand the social phenomena of political discourse.**
- Content moderation at scale for social media platforms
- Understanding political narrative propagation

Objectives

1. **Granular Multimodal Extraction:** VLM-based transcription of code-mixed text with noise filtering
2. **Robust Text Normalization:** Comprehensive cleaning pipeline for internet text entropy
3. **PIXAI Architecture Development:** Novel cross-attention mechanism for visual-textual fusion.
 - a. Why not Unimodal?
 - b. Why not Early Fusion?
 - c. Why not Late Fusion?
4. **Real-Time Deployment:** Production-ready web API with asynchronous processing

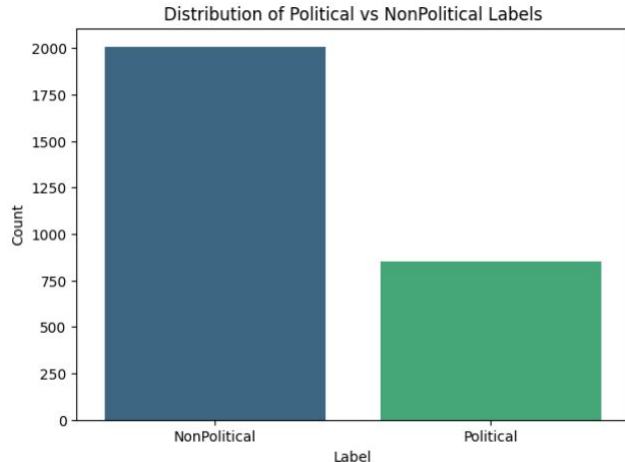
Dataset Overview

Meme Decode Dataset

- **Total Samples:** 2,860 code-mixed Bengali memes
- **Training Set:** 2,002 samples (70%)
- **Validation Set:** 429 samples (15%)
- **Test Set:** 429 samples (15%)

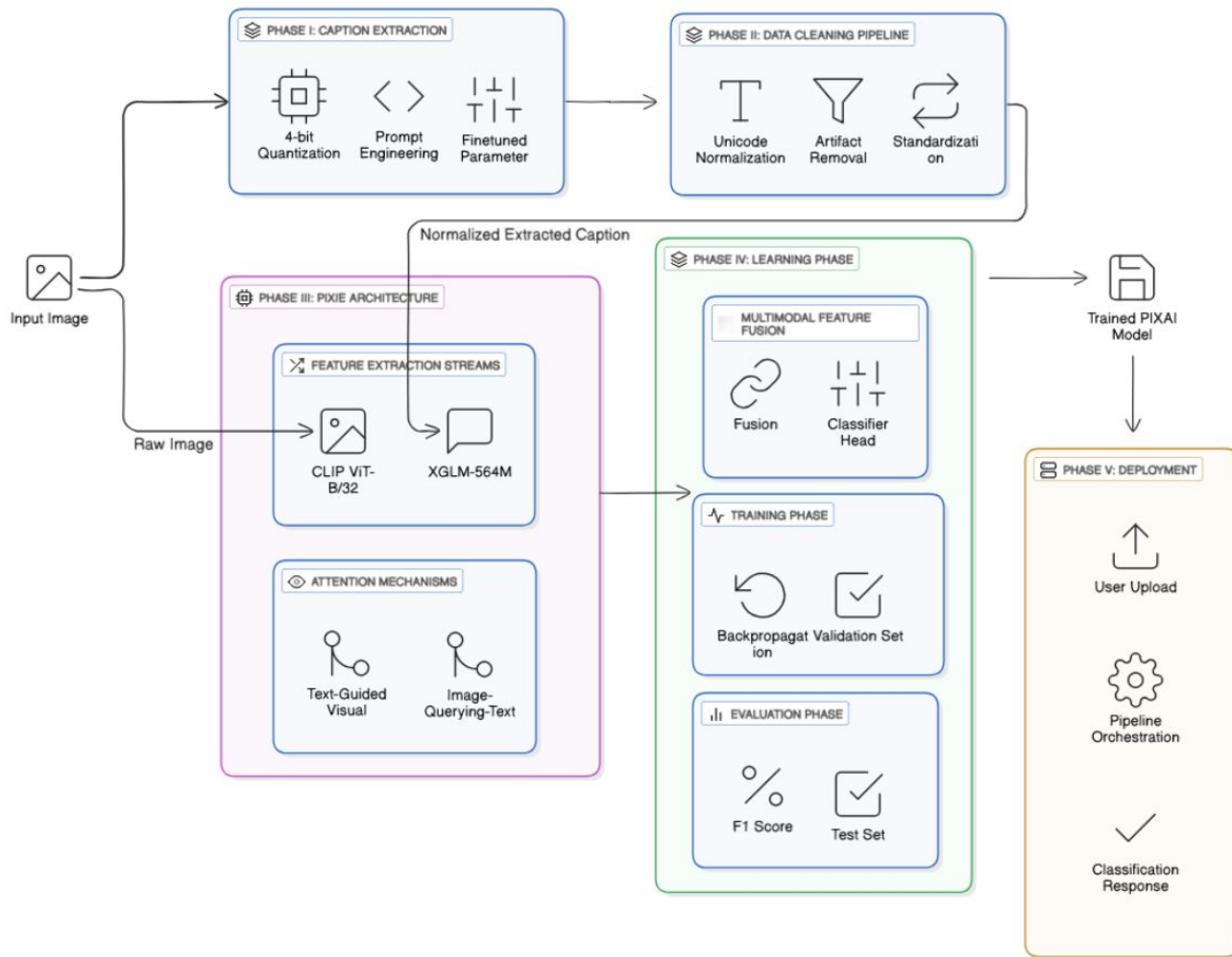
Class Distribution:

- Non-Political: 2,007 samples
- Political: 853 samples
- **Weighted loss ($\text{pos_weight} = 2007/853 = 2.35$)** applied to handle imbalance instead of other techniques (i.e. data augmentation, resampling etc) to keep the normal distribution.



System Architecture Overview

1. **Granular Multimodal Extraction:** VLM-based transcription of code-mixed text with noise filtering
2. **Robust Text Normalization:** Comprehensive cleaning pipeline for internet text entropy
3. **PIXAI Architecture Development:** Novel cross-attention mechanism for visual-textual fusion
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System Architecture - VLM Text Extraction

Traditional OCR (Tesseract):

- ***Dumb pattern recognition. No semantic understanding***
- Fails on mixed scripts, stylized fonts
- Cannot handle visual noise
- Terrible accuracy for memes, more suitable for low-noise clear documents

Our VLM Approach (Qwen/Gemini):

- **Semantic understanding.** Knows what it is extracting
- Accuracy: >90% extraction quality
- Context-aware correction of spelling errors
- Handles code-mixed text naturally
- Robust against stylized fonts and artifacts

System Architecture - Text Normalization Pipeline

Cleaning Internet Text Entropy

- Noise keyword filtering (watermarks, social media handles)
- Character mapping for stylized Unicode
- URL, emoji, and punctuation removal via regex
- NFKC Unicode normalization for Bengali script
- Whitespace standardization

Result: Clean, standardized input for robust classification

Hasan, T., Bhattacharjee, A., Samin, K., Hasan, M., Basak, M., Rahman, M. S., & Shahriyar, R. (2020).

Not Low-Resource Anymore: Aligner Ensembling, Batch Filtering, and New Datasets for Bengali-English Machine Translation.

EMNLP 2020.

System Architecture - Dual Cross-Attention Mechanism

- **Visual Stream:** CLIP (ViT-B/32) $\rightarrow$ 512D $\rightarrow$ 1024D projection (frozen). Works as static image feature extractor.
- **Textual Stream:** XGLM-564M $\rightarrow$ 1024D **contextual** embeddings.
- **Attention Stream I - Image-Querying-Text:**
 - Query: Visual Features
 - Key/Value: Textual Features
 - Function: This mechanism allows the model to identify which parts of the text are relevant to specific visual regions.
- **Attention Stream II - Text-Guided Visual Attention:**
 - Query: Visual Features
 - Key: Textual Features
 - Value: Visual Features
 - Function: Here, the text acts as a filter (Key) to re-weight the visual features (Value). This effectively “highlights” visual regions that are semantically aligned with the text.

Classification: Concatenated features (4096) $\rightarrow$ FC layers (32) $\rightarrow$ Sigmoid(Output) $\rightarrow$ Label

System Architecture - Dual Cross-Attention Mechanism (Cont'd)

$$\text{Attention}(Q_v, K_t, V_t) = \text{softmax} \left(\frac{Q_v K_t^\top}{\sqrt{d_k}} \right) V_t$$

- **Q (Query):** what the model is looking for
- **K (Key):** what the model can match against
- **V (Value):** information to be aggregated
- **Dot product:** measures similarity between queries and keys
- **Scaling:** stabilizes gradients for large dimensions
- **Softmax:** outputs a probability distribution that sums to 1.
- **Attention Weight:** signifies significance, multiplied with V to get a weight sum.

System Architecture - Training Hyperparameters

Parameter	Value
Epochs	5
Batch Size	16
Learning Rate	2×10^{-5}
Max Sequence Length	64 tokens
Optimizer	MADGRAD
Scheduler	Linear with Warmup
Loss Function	BCEWithLogitsLoss with pos_weight
Attention Heads (NUM_HEADS)	8
Attention Dimension (d_model)	1024
Attention Dropout	0.1
FC Layer Dropout	0.01

Table 6.1: Training Hyperparameters for PIXAI

System Architecture - Training Hyperparameters (Cont'd)

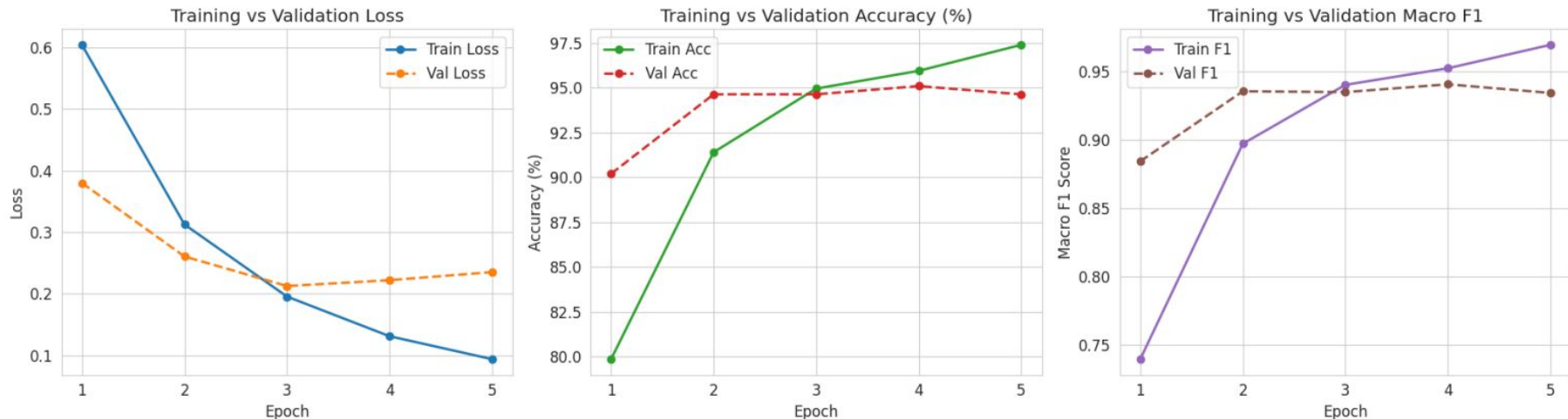


Figure A.1: (Training vs Validation) Loss, Accuracy and Macro F1

Notably, Epoch 4 achieves the highest validation Macro F1-score, confirming it as the optimal checkpoint and justifying its selection for final testing and deployment.

Result : Model Performance

Class	Precision	Recall	F1-Score	Support
Non-Political	0.9277	0.9801	0.9532	301
Political	0.9459	0.8203	0.8787	128
Accuracy			0.9324	429
Macro Avg	0.9368	0.9002	0.9159	429

Overall accuracy: 93.24% on a 429-sample test set.

Macro F1-score: 0.9159 confirms robustness despite class imbalance.

Non-Political class:

- Very high recall (**98.01%**)
- Neutral content is rarely misclassified or over-flagged

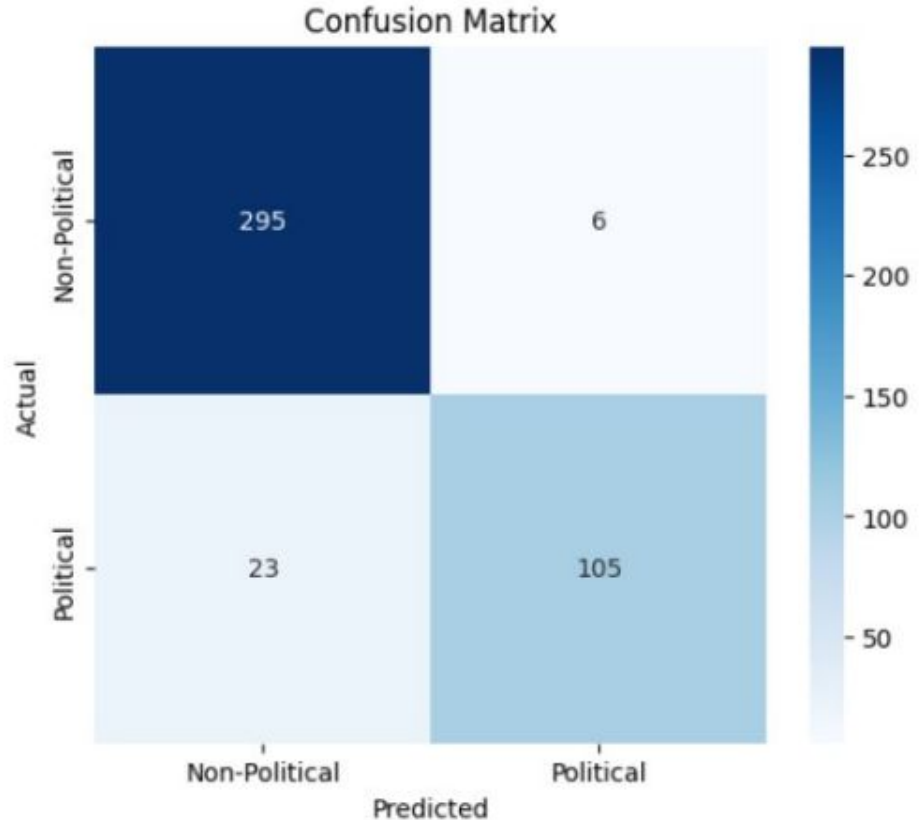
Political class:

- High precision (**94.59%**)
- When flagged as political, predictions are usually correct
- Lower recall (**82.03%**) reflects a **precision-first design choice**

Result : Model Performance (Cont'd)

- True Negatives: 295 (correctly identified non-political)
- True Positives: 105 (correctly identified political)
- False Positives: 6 (**minimal over-censorship**)
- False Negatives: 23 (**subtle satire detection challenge**)

Trade-off: High precision prioritized for safe deployment



System Architecture - Deployment of the Trained Model

Production-Ready FastAPI Backend

Features:

- Asynchronous request handling (async/await)
- 4-bit quantization for consumer GPU deployment
- Singleton pattern for model loading (lifespan context)
- Structured API contracts with Pydantic validation

Performance:

- Inference Latency: 2.5-4 seconds per image
- VRAM Usage: ~7GB with quantization
- Concurrent request support

System Integration Workflow

End-to-End Pipeline:

1. **User Upload:** Drag-and-drop image via web interface
2. **Text Extraction:** VLM-based OCR
3. **Normalization:** Clean and standardize extracted text
4. **Classification:** Trained PIXAI model inference with CLIP for the uploaded image + XGLM for the normalized extracted text.
5. **Response:** JSON output with confidence scores and probabilities

All orchestrated through FastAPI controller

Deployed System Screenshots

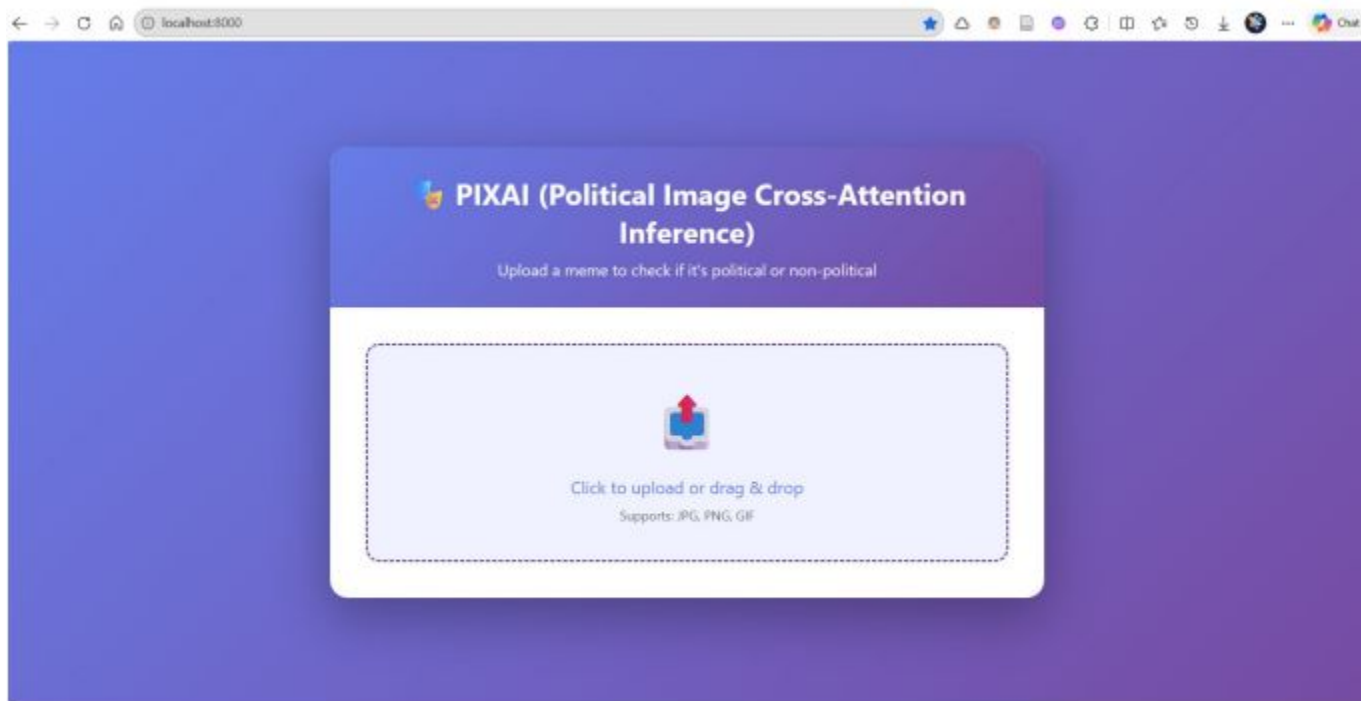


Figure 7.3: Webapp Interface

Correctly Classified *Political* Meme



Analyze Meme

Upload Another

This meme is **Political**

97.9%

97.9%

Political

2.1%

Non-Political

Extracted Text

Correctly Classified *Non-Political* Meme



Analyze Meme

Upload Another

This meme is **NonPolitical**

99.3%

0.7%

Political

99.3%

Non-Political

Extracted Text

Misclassified *Political* Meme

- Evolving political narratives, new symbols and slang emerge constantly.
- Subtle, implicit political cues embedded within context-dependent scenarios .
- Sarcasm and irony obscure intent.

Static models lack adaptability.

Lowering the threshold for *political* class might help.



Analyze Meme

Upload Another

This meme is **NonPolitical**

73.2%

26.8%

Political

73.2%

Non-Political

Key Contributions

1. First dedicated code-mixed political meme classifier for low-resource Bengali language.
2. Novel VLM-based extraction pipeline replacing traditional OCR
3. Dual cross-attention architecture for explicit multimodal reasoning
4. Production-ready deployment with asynchronous FastAPI backend
5. Established baseline: 93.24% accuracy, 0.9159 Macro F1

Limitations and Future Work

- Static modeling in dynamic political landscape
- Binary classification only (no satire/propaganda/sentiment detection)
- 2-4 second latency limits high-throughput scenarios
- Explainable AI with attention map visualization for trustworthiness.

Conclusion

PIXAI successfully demonstrates:

- Automated political meme classification for Bengali code-mixed content
- VLM-based extraction overcomes traditional OCR limitations
- Cross-attention fusion enables nuanced multimodal reasoning
- Real-time deployment achieves 93.24% accuracy with high precision

Significance: First computational framework for low-resource political discourse analysis in Bengali digital spaces.

Thank You. Questions?